Lecture 17: magnetism

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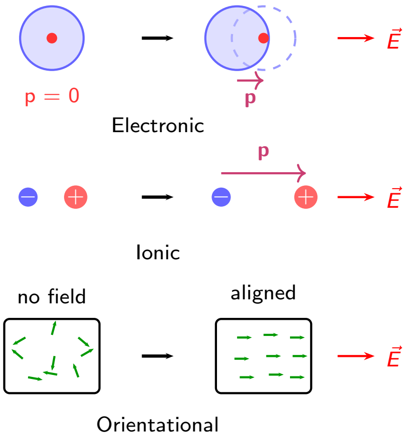
Remember: three polarization mechanisms in dielectrics

- **Electronic:** electron cloud displacement (all materials)
- **Molecular (ionic):** ion displacement in ionic crystals
- **Orientational:** permanent dipole alignment (polar molecules)
- Each has characteristic frequency cutoff



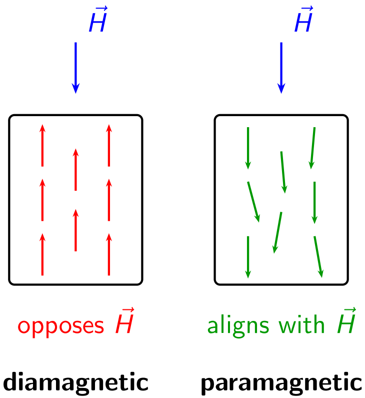
Remember: Maxwell's equations (general)

With the definitions $\mathbf{D} = \varepsilon \mathbf{E}$ ($\mathbf{D}$ = displacement field) and $\mathbf{B} = \mu \mathbf{H}$ ($\mathbf{H}$ = magnetic field),

$$\nabla \cdot \mathbf{D} = \rho$$
$$\nabla \cdot \mathbf{B} = 0$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$
$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Magnetism: two fundamental mechanisms

- All materials respond to magnetic fields, but through different mechanisms
- **Diamagnetism**: induced moments oppose applied field (negative response)
- **Paramagnetism**: pre-existing moments align with field (positive response)
- Understanding this distinction is key to all magnetic phenomena



Magnetic susceptibility: intrinsic materials response to magnetic field

- M = magnetization (magnetic dipole moment per unit volume)
- χ_m = magnetic susceptibility: $M = \chi_m H$

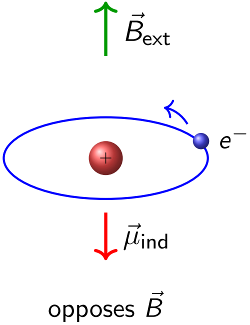
Fundamental relation

$$B = \mu_0(H + M) = \mu_0(1 + \chi_m)H$$

- **Diamagnets**: $\chi_m < 0$ (oppose field), typical $|\chi_m| \sim 10^{-5}$
- **Paramagnets**: $\chi_m > 0$ (enhance field), typical $\chi_m \sim 10^{-3}$–10^{-2}

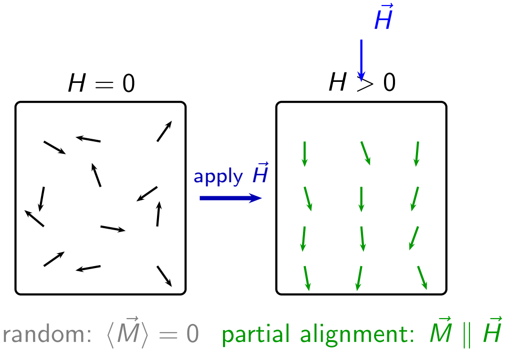
Diamagnetism: universal behavior but weak

- Diamagnetism is Lenz's law at atomic scale: electron orbits adjust to oppose flux change
- Present in **all** materials (there is always some orbital motion)
- Effect is weak: induced moment $\ll$ permanent atomic moments
- Examples: Cu, Ag, Au, Bi, most organic molecules ($\chi_m \sim -10^{-5}$)



Paramagnetism: aligning magnetic dipoles with the external field

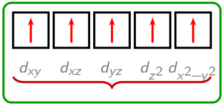
- Atoms with unpaired electrons have permanent magnetic moments
- Applied field partially aligns randomly oriented moments
- Effect dominates diamagnetism when unpaired electrons present
- Examples: O_2, Fe^{3+}, Mn^{2+}, rare earths ($\chi_m \sim 10^{-3}$)



Origin of the net magnetic moment: unpaired electrons

- Each electron has a spin ("mini current")
- Paired electrons (↑↓) cancel each other: zero net moment
- Only **unpaired** electrons contribute → net $\vec{\mu}$
- Transition metals: partially filled d shell → strongest paramagnets

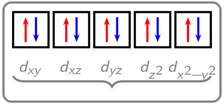
Fe^{3+}: [Ar] $3d^5$



paramagnetic

5 unpaired $\rightarrow \mu = 5.92\ \mu_B$

Zn^{2+}: [Ar] $3d^{10}$



diamagnetic

0 unpaired $\rightarrow \mu = 0$ (diamagnetic)

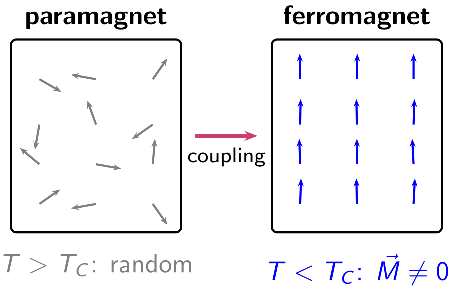
Summary: diamagnetism vs paramagnetism

- **Why is diamagnetism universal?**
  All electrons have orbital motion
- **Why is diamagnetism weak?**
  Induced moments $\ll$ permanent moments
- Key concept: induced (dia) vs pre-existing aligned (para)

	Dia	Para
χ_m sign	< 0	> 0
$\lvert\chi_m\rvert$	$\sim 10^{-5}$	$\sim 10^{-3}$
Origin	induced	pre-existing
Electrons	paired	unpaired

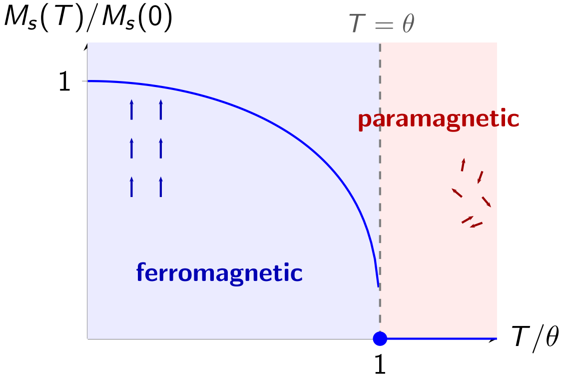
Ferromagnetism: spontaneous magnetization

- Ferromagnets exhibit net magnetization M even **without** external field H
- Spontaneous alignment of atomic magnetic moments below critical temperature
- Examples: iron (Fe), cobalt (Co), nickel (Ni), and their alloys
- Requires explanation beyond simple paramagnetic alignment



The Curie temperature θ

- **Above** θ: thermal energy overcomes internal field $\rightarrow$ paramagnetic
- **Below** θ: spontaneous magnetization exists without external field
- Phase transition: ferromagnetic $\leftrightarrow$ paramagnetic at $T = \theta$



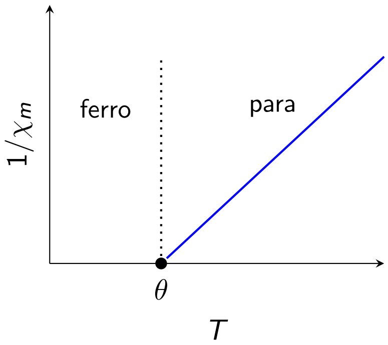
Susceptibility above the Curie temperature

For $T > \theta$: paramagnetic behavior with modified Curie law

Curie-Weiss law

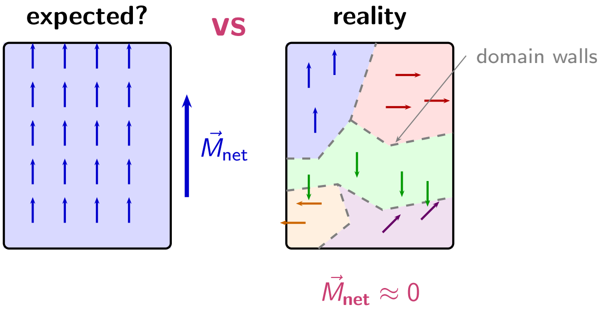
$$\chi_m = \frac{C}{T - \theta}$$

- Susceptibility **diverges** as $T \to \theta^+$
- Divergence signals onset of spontaneous ordering
- χ^{-1} vs T gives straight line with intercept at θ



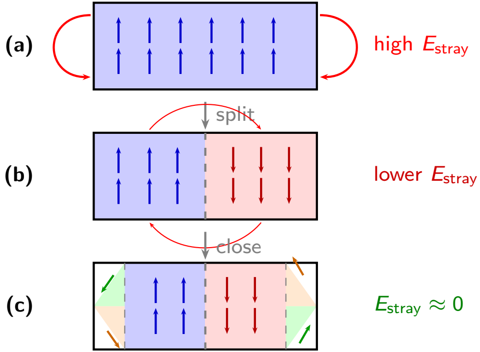
Ferromagnetic domains

- A ferromagnet is not uniformly magnetized in one direction
- It breaks into **domains**: regions of uniform $\vec{M}$, each pointing a different direction
- Domains can cancel $\to$ a ferromagnet may show **zero net magnetization**
- Why does this happen? $\to$ energy minimization (next slide)

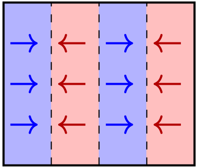


Why do domains form? $\rightarrow$ energy minimization

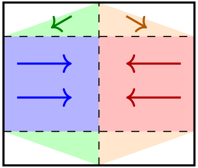
- Single-domain ferromagnet creates large external field $\rightarrow$ high magnetostatic energy
- Splitting into domains reduces external field energy
- Closure domains can eliminate stray field entirely
- Trade-off: domain walls cost energy, but field energy savings dominate



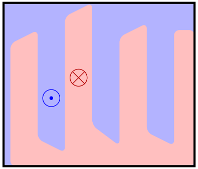
Domain patterns in real materials



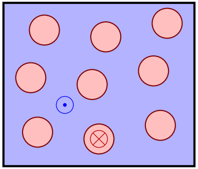
stripe domains
(uniaxial)



closure domains
(cubic, thick)



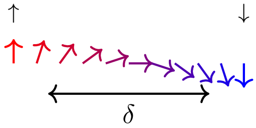
maze domains
(thin film, $\perp$)



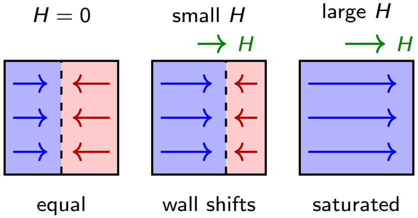
bubble domains
(thin film + H)

Domain walls and domain wall motion

Bloch wall: gradual spin rotation across wall width δ

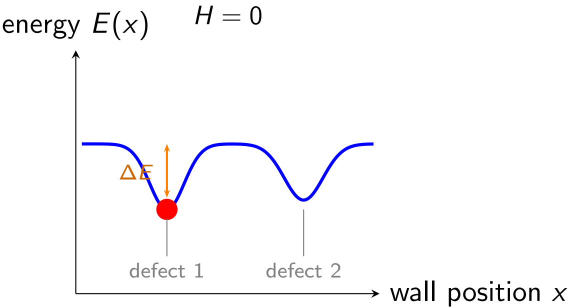


Wall motion under applied field H:



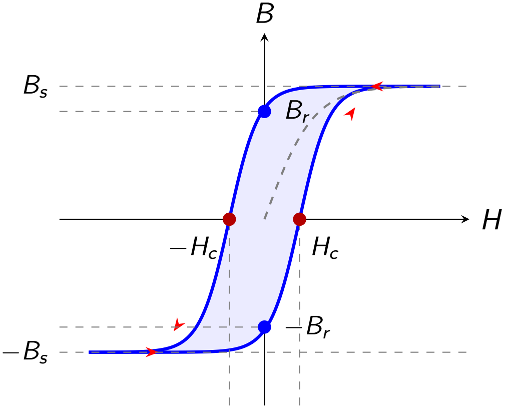
Irreversible pinning: defects trap domain walls

- Crystal defects (dislocations, grain boundaries, inclusions) pin domain walls
- Applied field tilts the energy landscape $\rightarrow$ reduces forward barrier
- Once wall jumps past defect, reverse barrier is large $\rightarrow$ **irreversible**
- This is the **microscopic origin of hysteresis**



The hysteresis loop in magnetization

- B_r (remanence): flux density when H returns to zero
- H_c (coercivity): reverse field needed to reduce B to zero
- B_s (saturation): all domains aligned with field
- Loop area = energy dissipated per cycle
- Loop shape reflects defect density and domain wall mobility



Energy dissipation in magnetic materials: loop area equals loss per cycle

- Work done per unit volume in one cycle: $W = \oint H\,dB$
- This equals the area enclosed by the hysteresis loop
- Energy dissipated as heat due to irreversible domain wall jumps

Energy dissipated per cycle

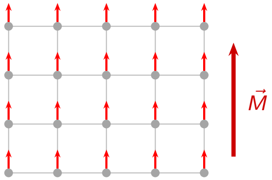
$$E_d = \oint H\,dB = \text{area of hysteresis loop}$$

Ferromagnetic vs antiferromagnetic exchange

Exchange energy between neighboring spins:

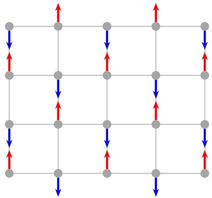
$$E_{\text{ex}} = -2J_{\text{ex}}\,\mathbf{S}_1 \cdot \mathbf{S}_2$$

- $J_{\text{ex}} > 0$: favors **parallel** spins (ferromagnetic)
- $J_{\text{ex}} < 0$: favors **antiparallel** spins (antiferromagnetic)
- Fe, Co, Ni: $J_{\text{ex}} > 0 \rightarrow$ ferromagnets
- Cr, Mn oxides: $J_{\text{ex}} < 0 \rightarrow$ antiferromagnets



net $\vec{M} \neq 0$

ferro ($J_{\text{ex}} > 0$)



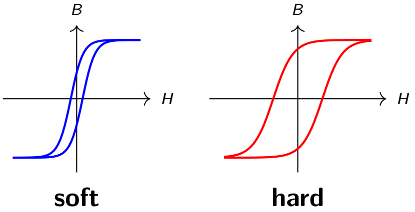
net $\vec{M} = 0$

antiferro ($J_{\text{ex}} < 0$)

Soft vs hard magnets: design requirements

- **Soft magnets**: high M_s, low H_c for easy switching
- **Hard magnets**: high B_r, high H_c to resist demagnetization
- Application determines material choice

Property	Soft	Hard
H_c	low	high
B_r	—	high
M_s	high	—
Use	transformers	permanent



Soft magnetic materials: selection criteria

- **High** M_s: maximizes flux density in transformer cores and motor laminations
- **Low** H_c: minimizes hysteresis loss per cycle
- **High electrical resistivity**: reduces eddy current losses at high frequencies

$$E_d \propto \oint H\,dB \quad \text{(proportional to loop area)}$$

- **Silicon steel**: power transformers (50/60 Hz)
- **Ferrites**: RF applications (high resistivity)
- **Amorphous alloys**: high-efficiency, low-loss cores

Concepts on soft vs hard magnet selection

1. What kind of applications need soft magnets? What about hard magnets?
2. What are the materials parameters you would choose when designing for magnets?
3. Why does higher $(BH)_{\mathrm{max}}$ enable smaller permanent magnets?

What we did not cover (there is much more!)

More magnetic ordering

- Ferrimagnetism (Fe_3O_4, garnets)
- Spin glasses and frustrated magnets

Anisotropy & micromagnetics

- Magnetocrystalline anisotropy
- Shape anisotropy

Quantum & many-body

- Angular momentum
- Spin–orbit coupling
- Kondo effect

Characterization

- NMR/neutron diffraction
- Mössbauer spectroscopy

Transport & devices

- Giant magnetoresistance (GMR)
- Tunnel magnetoresistance (TMR)
- Spintronics and spin-transfer torque
- Magnetic recording & hard drives
- MRAM

Modern frontiers

- Magnetic skyrmions
- Topological magnets
- Multiferroics
- 2D magnetism (CrI_3, Fe_3GeTe_2)